

Predicting the long-term impact of AI assistants on collective intelligence

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Abstract

Recent advances in artificial intelligence (AI) have enabled widespread use of AI assistants for various cognitive tasks, ranging from simple everyday tasks to scientific discovery and creative writing. While AI assistance can improve the performance of each individual, it can weaken their subsequent performance without AI and further limit the diversity of interests and opinions. These risks raise the question of whether human collective intelligence can persist as our society becomes increasingly reliant on AI assistants. To address this question, we develop an evolutionary model of collective intelligence, where each individual participate in a prediction task, consult AI, and imitate each other's prediction strategy. We find that "Chatbot" AI, which depends on users' interests, can prevent the emergence of collective intelligence by either limiting diversity or propagating systematic bias through overreliance. Directly incentivizing or penalizing AI use generally worsens these outcomes. Instead, a balanced incentive structure that preserve minority expertise and reward contributions to collective outcome can promote the emergence of collective intelligence. However, the inherent ability of the population's to make prediction in the absence of AI remains suboptimal even under this balanced incentive structure, raising concerns for continued AI reliance. **[SWP: conclusion?]**

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability (Galton, 1907; Woolley et al., 2010; Yaniv and Milyavsky, 2007; Hommes et al., 2005; Lorge et al., 1958). This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups (Dreyer et al., 2025; Simons, 2004; Berdahl et al., 2013) and human societies (Woolley et al., 2010; Riedl et al., 2021) to machine-learning systems (Breiman, 1996, 2001). Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences (Mann and Helbing, 2017), evolutionary biology (Wang et al., 2025), and computer science (Werfel, 2026). Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes (Cui and Yasseri, 2024).

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks (Noy and Zhang, 2023; Brynjolfsson et al., 2025), including scientific discovery (Lu et al., 2026) and creative writing (Doshi and Hauser, 2024). In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing (Bick et al., 2026; Sajadieh et al., 2026), the potential long-term impact of AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise (Vaccaro et al., 2024), and opinion aggregation (Burton et al., 2024; Tessler et al., 2024). Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions (Fügener et al., 2021; Doshi and Hauser, 2024; Lee and Chung, 2024; Meincke et al., 2025), a key requirement for the emergence of CI (Hong and Page, 2004; Lorenz et al., 2011; Aplin et al., 2014; Becker et al., 2017).

Our ability to predict the long-term impact of AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity (Bousquette, 2026; Keary, 2026). On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence (Thorp, 2023; Chawla, 2026). Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual participates in a prediction task from partial observations of under-

69 lying causal factors (Mann and Helbing, 2017; Wang et al., 2025; Hong et al., 2012).
 70 In particular, we allow each individual to independently consult with AI assistants
 71 and revise their predictions accordingly. The resulting model allows us to investi-
 72 gate the robustness of human CIs under different assumptions about the AI model’s
 73 ability to make predictions as well as incentive structures that are associated with
 74 the prediction task. In doing so, we assess the impact of directly incentivizing or
 75 penalizing AI uses on CI and further suggest a balanced incentive structure that can
 76 maintain CI in the presence of AI.

77 Mathematical model

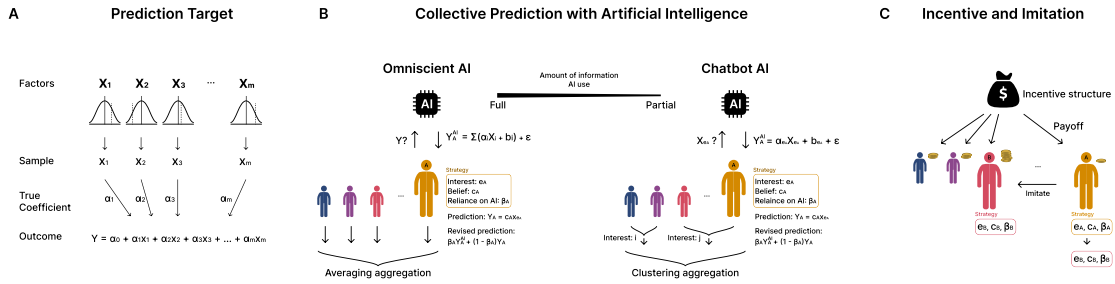


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

78 We begin by extending previously established models of CI, in which a group of
 79 players participates in a prediction task (Figure 1): each player makes an individ-
 80 ual prediction based on partial observations of the underlying causal factors, and
 81 the group forms a collective prediction by aggregating these individual predictions
 82 (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025). Specifically, the
 83 prediction target, Y , is modeled as a linear combination of M causal factors x_i , with
 84 fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from inde-
 85 pendent normal distribution (Figure 1A). Then, each individual A observes a single
 86 factor of their interest x_{e_A} and makes a prediction based on their own belief, c_A :
 87 $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then independently consult
 88 their AI assistants and revise their prediction based on their reliance on AI, β_A :
 89 $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are then aggregated

using two alternative aggregation rules, averaging and clustering, to form collective prediction.

To model AI, we assume that AI has already been trained on similar tasks and is able to make predictions based on partial or full observations of causal factors (Figure 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on its internal parameters, which can include factor-specific bias (b_i) and stochastic error ϵ . Here, we consider two extreme types of AI response, which differ in the amount of information AI uses when making predictions. In one extreme, the AI can observe all causal factors and use all information to make predictions, regardless of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works like a chatbot, where the AI predicts the outcome based only on the user’s interest: $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’ and ‘Chatbot AI’, respectively.

We assume that appropriate aggregation rule for collective prediction depends on the type of AI. When Chatbot AI is used, each revised prediction depends on a single factor. Thus, we use clustering aggregation, under which revised predictions are first averaged among individuals who are interested in the same factor, and then summed across factors. When Omniscient AI is used, however, each revised prediction contains information about all causal factors, in which case clustering these predictions by interest would be inappropriate. Instead, we use averaging aggregation, under which revised individual predictions are averaged directly to form the collective prediction.

Each generation consists of multiple round of predictions, and each player is rewarded a payoff according to the incentive structure (Figure 1C). For simplicity, we assume that a sufficiently large number of prediction rounds occurs within each generation, allowing us to use the expected payoff value rather than computing the payoffs for every round. We consider two incentive structures from previous work, “feedback” and “niche-expert”, which have been shown to promote near-perfect collective accuracy in the absence of AI (Wang et al., 2025). The feedback structure gives higher payoff to individuals who can push the collective prediction toward the true target, regardless of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In contrast, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A} denotes the proportion of players interested in the same factor, thereby allowing the niche-expert structure to preserve interest diversity. We note collective intelligence fails to emerge under the combination of averaging aggregation with niche-expert structure in the absence of AI (Wang et al., 2025).

Through time, the group evolves by allowing each individual to imitate peers who received higher payoffs (Figure 1C). Specifically, after each generation, two individuals are randomly sampled, and one imitates all three strategies (the factor of interest, factor belief, and reliance on AI) of the other with a probability determined by the difference in their expected payoffs (Materials and Methods). Individual predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation

133 rules. Collective accuracy is defined as the reduction in prediction error achieved
 134 by the collective prediction relative to a baseline prediction α_0 (Wang et al., 2025):
 135 $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective
 136 accuracy reaches a near perfect level.

137 This model now allows us to explore the impact of AI on CI depending on the
 138 type of AI and the incentive structure. Here, we begin by comparing the emergence
 139 of CI with and without AI and subsequently investigate a how different incentive
 140 structures affect the emergence of CI in the presence of AI.

141 AI can either promote or prevent the emergence of 142 CI

143 To understand the impact of AI, we first compare collective dynamics with and
 144 without AI. First, consistent with Wang et al. (2025), simple imitation rules and
 145 incentive structures allow collective intelligence to emerge in the absence of AI (Fig.
 146 2A,C,D). One exception is the use of simple averaging with niche-expert incentive
 147 structures, under which the collective accuracy of the population remains near zero.
 148 This is because enhancing individual accuracy causes individuals to predict close to
 149 the true coefficient of their interest, but simply averaging these predictions does not
 150 necessarily produce an accurate estimate of the full prediction target (Fig. S1).

151 We find that AI usage can have heterogeneous outcomes depending on its type.
 152 For example, the use of Omniscient AI can either preserve CI under the feedback
 153 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
 154 expert incentive structure (Fig. 2B). These results are robust across different levels
 155 of AI accuracy and can be shown analytically by assuming a sufficiently large group
 156 of players (Fig. S2; see SI Evolutionary dynamics section). **[SWP: need to get**
 157 **back.]**

158 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
 159 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
 160 structure, the use of Chatbot AI leads to a loss of diversity in the population's
 161 interest, making it impossible to account for effect of some causal factors (Fig. S2,
 162 S3). This causes an increase in collective variance and stochastic decrease in collective
 163 accuracy (Fig. 2C; Fig. S3). We note that the loss of interest diversity may arise
 164 even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot
 165 AI makes the group more vulnerable to such loss (Fig. S4). Under the niche-
 166 expert incentive structure, the population becomes overly reliant on AI, causing the
 167 collective prediction to inherit the AI's bias (Fig. S3). This leads to an increase in
 168 collective bias and a decrease in collective accuracy (Fig. 2; Fig. S3). Interestingly,
 169 we find that the niche-expert structure can prevent the use of AI under specific
 170 assumptions about the relationship between the true coefficient and AI bias (Fig.
 171 S2). In this case, the population reduces **[JS: I think it is difficult to say that it**
 172 **completely excludes AI, because reliance on AI does not approach zero. This may be**

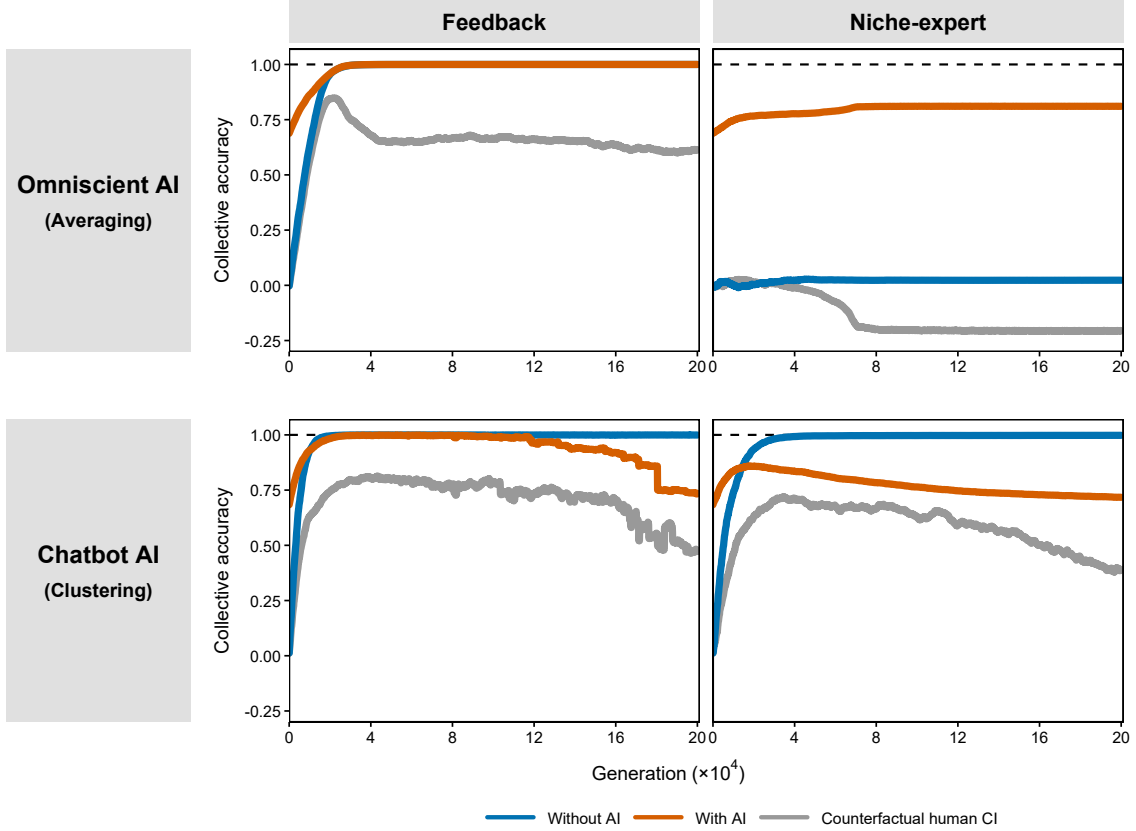


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Blue lines represent trajectories of collective accuracy without AI assistance. Orange lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

173 *due to the limited population size.* the reliance on AI assistance, allowing the CI to
 174 emerge eventually. However, this phenomenon can substantially delay the emergence
 175 of CI (Fig. S5).

176 To investigate whether the group can retain its predictive ability when AI assis-
 177 tance suddenly becomes unavailable at any given time, we quantify counterfactual
 178 collective accuracy based on unrevised predictions. Across all scenarios, we find that
 179 the counterfactual collective accuracy is lower than the collective accuracy of the

180 corresponding population (Fig. 2).

181 In summary, an omniscient AI can promote CI, but AI operating with limited
 182 information may prevent the emergence of CI. Moreover, reliance on AI can limit
 183 inherent ability of the population to make accurate predictions on their own, raising
 184 concerns about future AI usage. These findings further underscore the need for in-
 185 centive structures that can mitigate the potential risks associated with AI assistance.

186 Directly incentivizing or penalizing AI use can ad- 187 ditionally harm CI

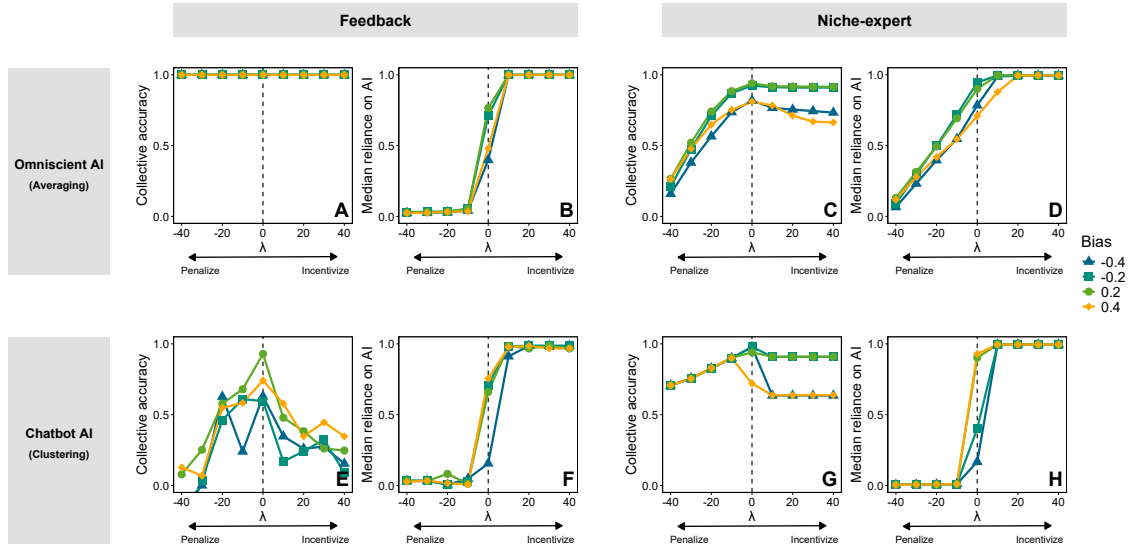


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

188 Given the strong influence of AI on collective outcomes, a straightforward inter-
 189 vention is to directly incentivize or penalize the use of AI. To investigate whether

such direct incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$), and β_A represent individual reliance on AI. [SWP: maybe we need to call incentive pi earlier on...] [SWP: also good to define beta again here as a reminder.]

Directly incentivizing or penalizing AI use can further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions can push the group to always [JS: wholly?] ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B, D, F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but this represents a narrow parameter regime (Figure 3G).

Overall, these results indicate that simply incentivizing or penalizing the AI, independent of individuals' performance, use may exacerbate rather than resolve the risk of AI on CI. Thus, we also considered an incentive structure where interventions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.] [JS: Should I add the multiplicative intervention to the SI? I remember that we didn't reach a clear conclusion during our last meeting about whether to include it.]

Balanced incentives mitigate AI-induced failures of CI

Neither the original incentive structures nor simple interventions targeting AI use are sufficient to promote CI. What incentive structure, then, would work in the presence of AI? Here, we propose a "balanced" incentive structure, modeled as the average of the feedback and niche expert incentives.

The core idea behind the balanced structure is to maintain interest diversity and prevent collective bias. On one hand, the feedback incentive structure is prone to the loss of interest diversity but can prevent collective bias by incentivizing players whose predictions lie opposite to the collective prediction. On the other hand, the niche-expert structure can maintain interest diversity because it directly incentivizes minority interests. However the niche-expert structure cannot prevent collective bias because, within the same factor of interest, individuals are rewarded only according to their individual predictive performance, which in turn homogenize the predictions.

Therefore, when the two incentive structures are combined, the population achieves CI by maintaining interest diversity while limiting collective bias (Fig. 4A, S6). Un-

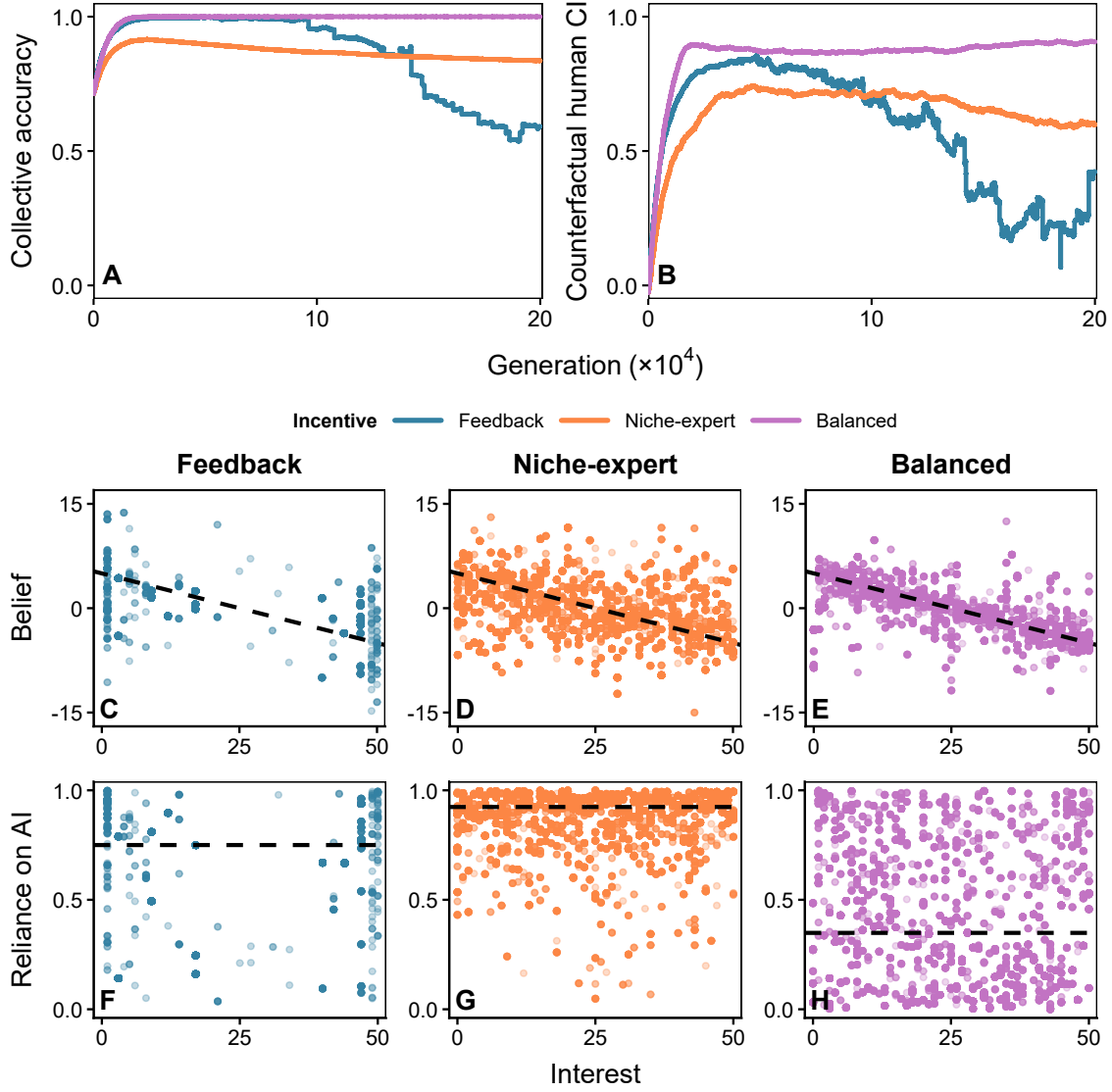


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

229 der the balanced structure, individuals remain more evenly distributed across causal
 230 factors, and their beliefs lie closer to the true coefficients than under either incentive

231 structure alone (Fig. 4C–E). The balanced structure also produces higher counter-
232 factual collective accuracy than the feedback or niche-expert structure (Fig. 4B,
233 S6). These patterns are robust to how the two incentive structures are weighted
234 when combined (Fig. S7), and are driven by the emergent property of the balanced
235 structure, under which the population increases its reliance on AI as AI becomes
236 more accurate (Fig S8).

237 *[SWP: may need some robustness check rather than a case study...]*

238 Discussion

239 The ongoing race towards the development of “better” AIs continues to pose ques-
240 tions about the societal benefits and risks of AI uses (Bender et al., 2021; Burton
241 et al., 2024; Cui and Yasseri, 2024). Here, we used mathematical models of evolu-
242 tionary dynamics to investigate how AI uses and incentive structures jointly affect
243 the emergence of CI, which is defined as the population’s ability to make accurate
244 predictions. We find that an “omniscient” AI model, which incorporates information
245 about all relevant causal factors, can improve collective accuracy, whereas a “chat-
246 bot” AI model, whose outcome depends on the user’s interest, can impair it. These
247 contrasting outcomes arise because AI assistance affects not only individual predic-
248 tions, but also the population’s interest diversity, beliefs, and reliance on AI. We also
249 show that directly incentivizing or penalizing AI uses, independent of individual’s
250 performances, can further limit the collective accuracy of the population. Instead,
251 a balanced incentive structure can maintain CI by promoting diversity and limiting
252 bias.

253 Our analysis reveals that AI can prevent the emergence of CI through two differ-
254 ent mechanisms. First, AI use can cause individuals to concentrate on a subset of the
255 factors that determine the prediction target, thereby reducing diversity in their inter-
256 ests. This result is broadly consistent with recent empirical studies showing that AI
257 assistance can improve individual performance but limit the diversity of outputs, in
258 domains ranging from scientific research (Hao et al., 2026) to creative writing (Doshi
259 and Hauser, 2024). Second, over-reliance on AI can cause the population to inherit
260 inherent biases in AI’s predictions (Fügener et al., 2021; Vicente and Matute, 2023),
261 causing the prediction accuracy to decrease eventually. This mechanism can prevent
262 the emergence of CI even when the interest diversity is maintained because biases in
263 AI predictions can be amplified when individual predictions are aggregated. Thus,
264 preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI
265 uses need to be calibrated to prevent systematic AI biases from becoming collective
266 biases (Bućinca et al., 2021).

267 Our analyses of counterfactual collective accuracy also raises an important con-
268 cern for future AI uses: while AI uses may improve the apparent performance of
269 individuals and, in some cases, the population as a whole, it can simultaneously re-
270 duce the population’s ability to make accurate predictions without AI. These results

271 echo findings from a recent field experiment, which demonstrated that students,
 272 who seemingly performed better in solving mathematics problems under AI assis-
 273 tance, performed worse on their own than those who did not have access (Bastani
 274 et al., 2025). These findings highlight the need to evaluate AI tools not only by
 275 the performance gains they provide during use, but also by whether individuals and
 276 populations retain the ability to perform without them (Melumad and Yun, 2025;
 277 Ke et al., 2026). Making this distinction clear will be particularly important for
 278 education, where AI may help students complete a task without helping them learn
 279 how to do it on their own (Bastani et al., 2025; Melumad and Yun, 2025).

280 As generative AIs and large language models are already generating substantial
 281 benefits, many organizations, both industrial and academic, are increasingly seek-
 282 ing to improve their collective performance by regulating AI uses (Thorp, 2023;
 283 Bousquette, 2026; Chawla, 2026; Keary, 2026). Such regulations may take direct
 284 forms, including AI-use policies or token-based leaderboards, or may operate indi-
 285 rectly through other incentives, such as productivity rewards or the social stigma
 286 associated with AI use (Reif et al., 2025). Our model suggests that directly incen-
 287 tivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on
 288 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
 289 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
 290 venting groups from using AI even when it performs well in the relevant domain.
 291 Our result instead shows that appropriate AI use and accurate CI can be promoted
 292 by focusing on the traditional properties of groups that have been emphasized for
 293 decades: maintaining diversity and preventing collective bias, rather than on AI use
 294 itself (Hong and Page, 2004; Lorenz et al., 2011; Mann and Helbing, 2017).

295 There are several limitations to our analyses. First, our model allows for social
 296 learning among humans, but AI’s inherent ability remains fixed throughout the evo-
 297 lutionary process. In reality, AI systems can also learn from human-generated data
 298 (Ouyang et al., 2022). Such bidirectional learning increase the similarity between
 299 human and AI errors and therefore further reduce the group performance (?). Sec-
 300 ond, we model AI errors using fixed, factor-specific additive biases and independent
 301 random errors. This assumption allows random errors to be reduced through ag-
 302 gregation, while systematic biases can still persist in the collective prediction. In
 303 reality, AI errors may have more complex structures, including biased estimates of
 304 factor effects and correlated errors across individuals, factors, or repeated interac-
 305 tions. Further empirical and theoretical work is therefore needed to understand how
 306 these error structures affect the emergence of CI. Finally, following established mod-
 307 els of CI, we assume that the prediction target is a linear combination of independent
 308 causal factors. This assumption does not hold for many real-world tasks, where causal
 309 factors may be correlated or interact nonlinearly. Future studies should therefore in-
 310 vestigate how nonlinear relationships between causal factors and prediction targets
 311 affect the interaction between AI assistance and the emergence of CI.

312 The rapid adoption of AI assistants is changing how individuals learn, work, and
 313 make decisions (Tessler et al., 2024; Glickman and Sharot, 2025; Sajadieh et al.,

2026; Bick et al., 2026; Ke et al., 2026). While the immediate benefits and risks of these systems have been evaluated at the individual level Noy and Zhang (2023); Bastani et al. (2025), our study raises a broader concern that their widespread AI use may reshape collective intelligence of human society. Our results further underline the importance of designing reward structures that preserve minority expertise and prevent AI biases from persisting across the population. A detailed empirical understanding of how AI assistance reshapes diversity, expertise, and reliance at the population level will therefore be critical for predicting the dynamics of future of human collective intelligence.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \dots + \alpha_M X_M, \quad (2)$$

where the coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and makes a prediction by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player’s prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player’s strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

Second, when the group is assisted by the Chatbot AI, both the player's initial prediction and the AI's prediction are based on the same causal factor. Each revised

prediction therefore contains information about only one factor and can be treated as an estimate of that factor’s contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs based on the underlying incentive structure. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A ’s revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

Previous research has shown that these two incentive structures can produce near perfect collective accuracy through social learning, except when the niche-expert incentive is combined with simple averaging (Wang et al., 2025). They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

To examine the effects of directly encouraging or discouraging AI use, we modify the feedback and niche-expert structures by adding a term proportional to the

415 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

416 Here, λ represents the strength of the direct incentive or penalty applied to AI use. A
 417 positive value of λ rewards AI reliance, whereas a negative value penalizes it. When
 418 $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to
 419 the original feedback and niche-expert structures.

420 The balanced structure is defined as the arithmetic mean of the feedback and
 421 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

422 Simulations

423 All simulations are conducted using an agent-based model with 50 causal factors
 424 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 425 target are sampled at the beginning of the simulations and subsequently held fixed.

426 Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the
 427 simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization.
 428 Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

429 Players’ initial strategies are also sampled from predefined probability distribu-
 430 tions. The distribution of individual belief c_A depends on the aggregation rule. Under
 431 simple averaging, each factor’s contribution is scaled by the proportion of players in-
 432 terested in that factor. We therefore sample beliefs from a Gaussian distribution
 433 with mean zero and standard deviation 100, following the scaling used in previous
 434 work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a
 435 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 436 are averaged separately within each interest group and then summed, no additional
 437 scaling of c_A is required. The initial interest of each player is sampled uniformly
 438 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 439 distribution on $[0, 1]$.

440 We assume that prediction events occur on a substantially faster timescale than
 441 social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$
 442 of each player analytically rather than explicitly sampling prediction events. The
 443 expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the
 444 stochastic AI error ϵ , conditional on the current population strategies and the fixed
 445 target parameters. Closed form expression of expected payoffs are derived for all
 446 incentive structures and are presented in the Supplementary Methods.

447 Each generation consists of an expected payoff calculation followed by one op-
 448 portunity for pairwise social learning. Two distinct players, A and B , are randomly

449 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

450 where s represents the strength of social learning and was set to 50 in all simulations.
 451 We introduce a mutation rate μ , such that with probability μ , player A does not imi-
 452 tate player B but instead redrew all strategy components from the same distributions
 453 used to initialize the population. The mutation rate is set to zero in all simulations
 454 except those presented in Supplementary Fig. S4 **[SWP: need to check]**.

455 Unless otherwise specified, each simulation is run for 200,000 generations. When a
 456 result is represented by a single summary value rather than as a temporal trajectory,
 457 the metric was averaged over the final 10,000 generations.

458 To evaluate the dynamics of collective prediction, we define six summary met-
 459 rics: collective accuracy, collective bias, collective variance, counterfactual collective
 460 accuracy, interest diversity, and median reliance on AI. At each population state,
 461 expectations and variances are evaluated analytically over the distributions of the
 462 causal factors and AI errors rather than estimated from sampled prediction events.
 463 Collective bias and collective variance constitute the bias–variance decomposition of
 464 the group level collective prediction error.

465 Collective accuracy is normalized relative to the baseline prediction α_0 . A nega-
 466 tive value indicates that the collective prediction performs worse than the baseline.
 467 Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the
 468 evolved interests and beliefs of the AI-assisted population. In calculating interest
 469 diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

470 Data and code availability

471 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
 472 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

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477 Contributions

478 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
 479 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
 480 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
 481 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

482 **Competing Interests**

483 The authors declare no competing interests.

Supplementary Text

Here, we derive the expected payoffs and collective accuracy used in the simulations. We consider Omniscient AI and Chatbot AI separately because they contain different amounts of information and therefore require different aggregation rules. We then analyze the corresponding evolutionary dynamics under a deterministic large-population approximation.

S1 Omniscient AI

We first consider Omniscient AI, which incorporates all causal factors regardless of the player's interest. We derive individual and collective predictions before calculating the expected payoffs under each incentive structure.

S1.1 Model

Because Omniscient AI observes all M causal factors, its prediction for player A is given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (\text{S1})$$

where b_i represent factor-specific biases and ϵ represents stochastic error, following a Gaussian distribution with mean zero and standard deviation τ . A new stochastic error is generated independently for each player and prediction event.

Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (\text{S2})$$

In contrast to the AI prediction, player A 's initial prediction depends only on the causal factor associated with their interest. Their initial and revised predictions are therefore given by:

$$Y_A = c_A X_{e_A} \quad (\text{S3})$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (\text{S4})$$

represents player A 's interest, c_A represents their belief about the effect of the corresponding factor, and β_A represents their reliance on AI. As in the main text, we set $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

508 S1.2 Aggregation

509 Because the AI component of each revised prediction contains information about all
510 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (S5)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (S6)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j}\} \quad (S7)$$

511 To separate the contributions of AI and the players' initial predictions, we define the
512 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
513 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S8)$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (S9)$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (S10)$$

514 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
515 remaining after averaging, and γ_i represents the contribution of players' initial pre-
516 dictions to factor i . Using these definitions, the revised collective prediction can be
517 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (S11)$$

518 Thus, the collective prediction consists of an AI-derived component, the contributions
519 of the players' initial predictions, and the stochastic AI error that remains after
520 aggregation.

521 S1.3 Expected payoffs

522 We next derive the expected payoff under each incentive structure. Expectations are
523 taken over the distributions of the causal factors and stochastic AI errors, conditional
524 on the current population strategies and the fixed parameters of the prediction target.

525 Feedback

526 Under the feedback structure, a player's payoff depends on the alignment between
 527 their revised prediction and the residual error of the collective prediction. To express
 528 this residual, we first define the remaining error in the intercept and in the coefficient
 529 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S12})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S13})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S14})$$

530 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 531 remaining error associated with factor i . Because the causal factors and stochastic AI
 532 errors are independent and have zero mean, cross terms vanish when the expectation
 533 is taken. Substituting the revised individual prediction and collective residual into
 534 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S15})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S16})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^m \delta_i X_i - \bar{\eta} \right\} \quad (\text{S17})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S18})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S19})$$

535 The first term represents the alignment between the AI-derived prediction and the
 536 collective residual, whereas the second term represents the contribution of the player's
 537 initial prediction. The final term arises because the stochastic error in player A 's AI
 538 prediction also contributes to the collective prediction. This finite-population effect
 539 decreases as population size increases.

540 Niche-expert

541 Under the niche-expert structure, the payoff is the negative squared error of the
 542 revised individual prediction, weighted by the frequency of the player's interest.
 543 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S20})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S21})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S22})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S23})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S24})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S25})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S26})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)\alpha_0 + \beta_A B$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\}^2] \quad (\text{S28})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S29})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S31})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re})\right] + \lambda\beta_A, \quad (\text{S32})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2\right] + \lambda\beta_A. \quad (\text{S33})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

557 S1.4 Expected accuracy

558 Using the residual defined above, the mean squared error of the revised collective
559 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S34})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S35})$$

560 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S36})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S37})$$

561 The collective prediction error consists of the squared error in the intercept, the re-
562 maining error associated with each causal factor, and the stochastic AI error remain-
563 ing after averaging across players. Counterfactual collective accuracy is obtained
564 from the same expression by setting $\beta_A = 0$ for all players while retaining their
565 evolved interests and beliefs.

566 S2 Chatbot AI

567 Unlike Omniscient AI, Chatbot AI uses only the causal factor associated with each
568 player's interest. This causes the information contained in AI predictions to depend
569 directly on the distribution of interests in the population. Here, we derive the collec-
570 tive prediction, expected payoffs, and expected collective accuracy under this form
571 of AI assistance.

572 S2.1 Model

573 The prediction generated by Chatbot AI for player A depends on the causal factor
574 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S38})$$

575 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
576 The stochastic error follows a Gaussian distribution with mean zero and standard
577 deviation τ and is generated independently across players and prediction events. The

578 player's initial and revised predictions are given by Player A's initial and revised
 579 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S39})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S40})$$

580 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 581 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 582 tion remains specific to the single factor associated with the player's interest.

583 S2.2 Aggregation

584 Because both the player's initial prediction and the Chatbot AI prediction contain
 585 information about the same causal factor, we use clustering aggregation. Revised
 586 predictions are first averaged among players interested in the same factor and then
 587 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S41})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S42})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon], \quad (\text{S43})$$

588 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 589 players in that set.

590 To simplify the collective prediction, we define the effective coefficient, weighted
 591 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S44})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S45})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S46})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon, \quad (\text{S47})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S48})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S49})$$

where $\mathcal{I} = i \in 0, \dots, M > 0$ denotes the set of represented interests.

S2.3 Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S50})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S51})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S52})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S53})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S54})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S55})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S56})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

612 Niche-expert

613 Under the niche-expert structure, the negative squared error of each revised indi-
 614 vidual prediction is weighted by the frequency of the player's interest. We consider
 615 $e_A = 0$ separately because these players make constant predictions rather than factor-
 616 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S57})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S58})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S59})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S60})$$

617 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 618 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S61})$$

$$= -\mathbb{E} [\rho_{e_A} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} \quad (\text{S62})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S63})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S64})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S65})$$

619 In both cases, the expected squared error consists of the mismatch between the
 620 player's belief and the true coefficient, contributions from causal factors omitted
 621 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 622 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 623 interest.

624 Incentivizing or penalizing AI use

625 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 626 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S66})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S67})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S68})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S69})$$

Balanced

Under the balanced structure, the payoff corresponds to the arithmetic mean of the feedback and niche-expert payoff components:

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2} \right] \quad (\text{S70})$$

$$= \frac{\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - \mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right]}{2} \quad (\text{S71})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

S2.4 Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S72})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S73})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

643 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S74})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S75})$$

644 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
645 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

646 S3 Evolutionary dynamics

647 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
648 evolutionary dynamics using the replicator equation under both weak and strong
649 selections. We focus on the feedback and niche-expert structures, for which the long-
650 term dynamics can be characterized analytically. The direct interventions on AI use
651 and the balanced structure are examined through simulations.

652 Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff.
653 Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S76})$$

654 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an
655 internal stationary state, all strategies with positive frequency must have the same
656 expected payoff. A boundary state can also be stationary because a strategy with zero
657 frequency cannot increase unless it is introduced into the population. We therefore
658 assess the stability of boundary states by asking whether they can be invaded by a
659 rare alternative strategy.

660 S3.1 Omniscient AI

661 With Omniscient AI, every AI prediction contains information about the full predic-
662 tion target, and revised predictions are combined through averaging. As population
663 size increases, independent AI errors vanish from the collective prediction. How-
664 ever, these errors remain in individual prediction errors and can therefore continue
665 to affect niche-expert payoffs. We consider the two incentive structures separately.

666 S3.1.1 Feedback

667 Under the feedback structure, both the stochastic error in the collective prediction,
668 $\bar{\eta}$, and its contribution to expected payoff vanish as $N \rightarrow \infty$. Following previous

work (Wang et al., 2025), we define the Lyapunov function as the expected error of collective prediction:

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S77})$$

The function V is non-negative and equals zero when the collective prediction is identical to the target. To determine how V changes over time, we take its time derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S78})$$

$$= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S79})$$

$$= -2 \left[\left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S80})$$

For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S81})$$

Although K changes with the population state, it is common to all strategies at a given state and can therefore be treated as a constant with respect to the strategy index λ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S82})$$

Under the replicator equation, the rate of change in the population mean of a strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S83})$$

Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S84})$$

Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S85})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S86})$$

For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy λ is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda)c_\lambda G_{e_\lambda}. \quad (\text{S87})$$

This expected payoff is identical to the quantity inside the covariance above. Therefore,

$$\frac{dV}{dt} = -2\text{Cov}(u, u) = -2\text{Var}(u) \leq 0. \quad (\text{S88})$$

Because payoff variance is non-negative, the expected collective prediction error cannot increase and decreases whenever the strategies currently represented in the population differ in their expected payoffs. The derivative becomes zero when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition arises when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

S3.1.2 Niche-expert

Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each individual. Stochastic AI error therefore remains in the expected payoff even when $N \rightarrow \infty$, and we retain $\tau > 0$ throughout this analysis.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S89})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its

unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S90})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

To compare the optimized strategies across interests, let L_e denote the minimum expected individual squared error within interest e , $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S91})$$

If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating the expected squared prediction error at the corresponding boundary.

Interest 0 must be considered separately because $X_0 = 1$ and the player's belief determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S92})$$

Substituting this into the expected payoff gives a strictly concave function of β , with the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S93})$$

Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S94})$$

and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S95})$$

At this optimum, the player adjusts their belief to offset the systematic AI bias. The stochastic component of AI error cannot be removed in the same way and therefore limits the benefit of AI reliance.

We denote the optimal strategy within interest e by S_e . When $\beta_e^* = 1$, different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class. A stable state can contain only the optimal strategy class within each represented interest because a suboptimal strategy can be invaded by S_e .

741 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
742 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S96})$$

743 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equa-
744 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S97})$$

745 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S98})$$

746 This internal stationary state is locally stable. At equilibrium, all represented in-
747 terests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
748 interest increases slightly above its equilibrium value, its payoff decreases because
749 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an
750 interest decreases below its equilibrium value, its payoff increases. The replicator dy-
751 namics therefore generate a restoring force that reduces small deviations in interest
752 frequencies and drives the population back toward the internal stationary state.

753 If one or more interests are absent, the resulting boundary state can be stationary
754 under the replicator equation, but it is unstable against invasion. The payoff of a rare
755 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
756 population is negative when $L_e > 0$, so the rare interest increases when introduced
757 at sufficiently small frequency.

758 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
759 prediction strategies and their frequencies. Whether the constrained optimum has
760 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
761 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
762 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
763 interest frequency remain uniquely determined.

764 Having characterized this equilibrium, we next compare its collective prediction
765 with a prediction made entirely by AI. In particular, the expected squared error of
766 an Omniscient AI prediction is

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S99})$$

767 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
768 fore, because L_e is the minimum individual expected squared prediction error within
769 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S100})$$

Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S101})$$

Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S102})$$

The collective prediction is the average of individual revised predictions. By convexity of the squared error,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S103})$$

Therefore,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[(Y - Y^{\text{AI}})^2 \right]. \quad (\text{S104})$$

Thus, the diversity of optimized predictions allows the population to outperform an individual AI prediction at equilibrium.

However, outperforming AI does not imply perfect collective accuracy. Under averaging, the non-AI contribution associated with a particular factor is provided only by individuals interested in that factor. The predictions of individuals with other interests therefore attenuate that factor's coefficient in the collective prediction. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S105})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S106})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S107})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-expert structure preserves interest diversity and allows the population to outperform AI, but averaging prevents the population from reaching perfect collective accuracy.

Supplementary Figure

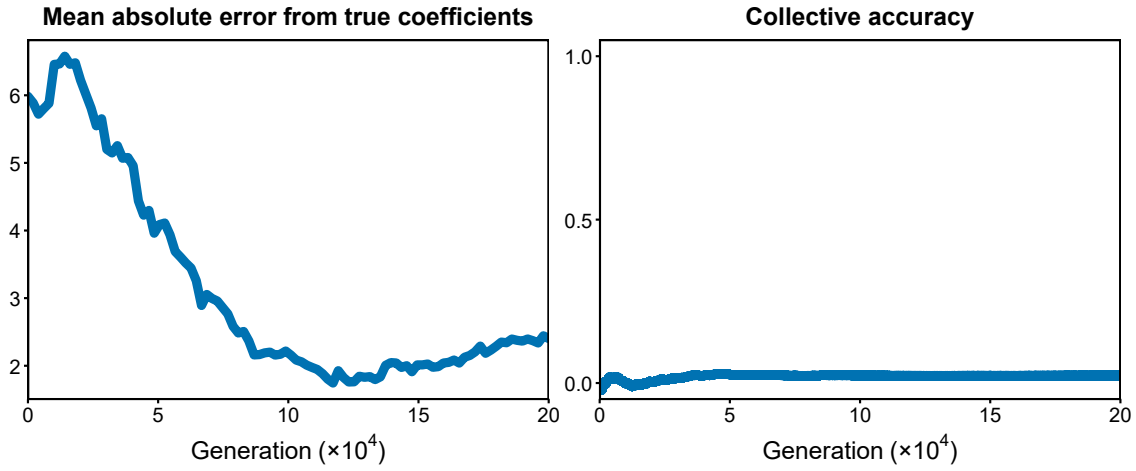


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $m + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

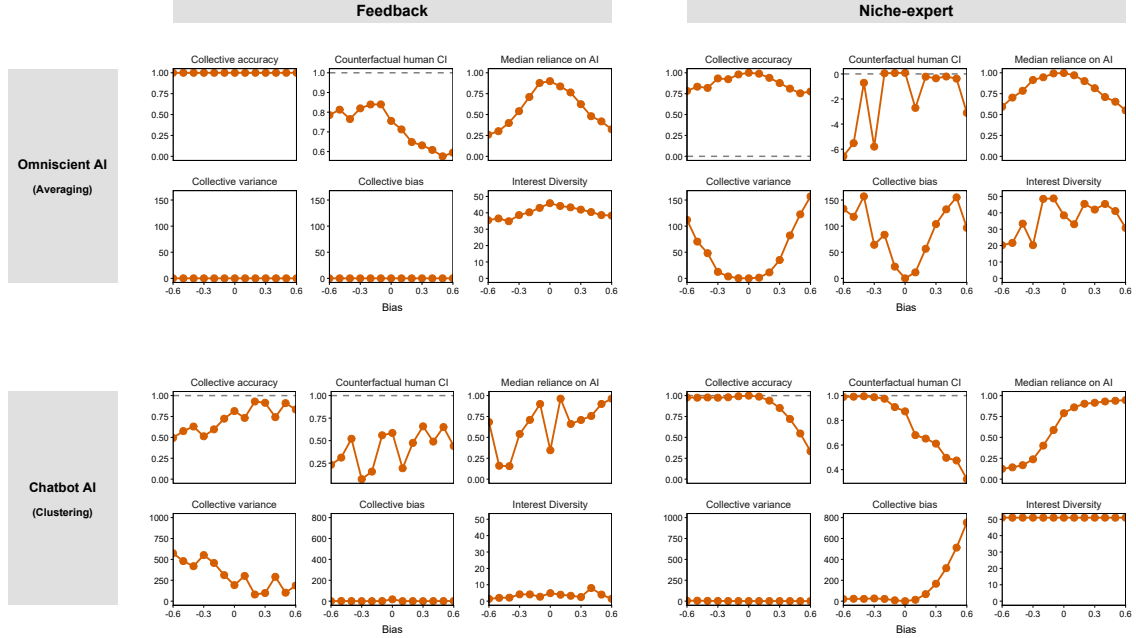


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (omniscient AI, and chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

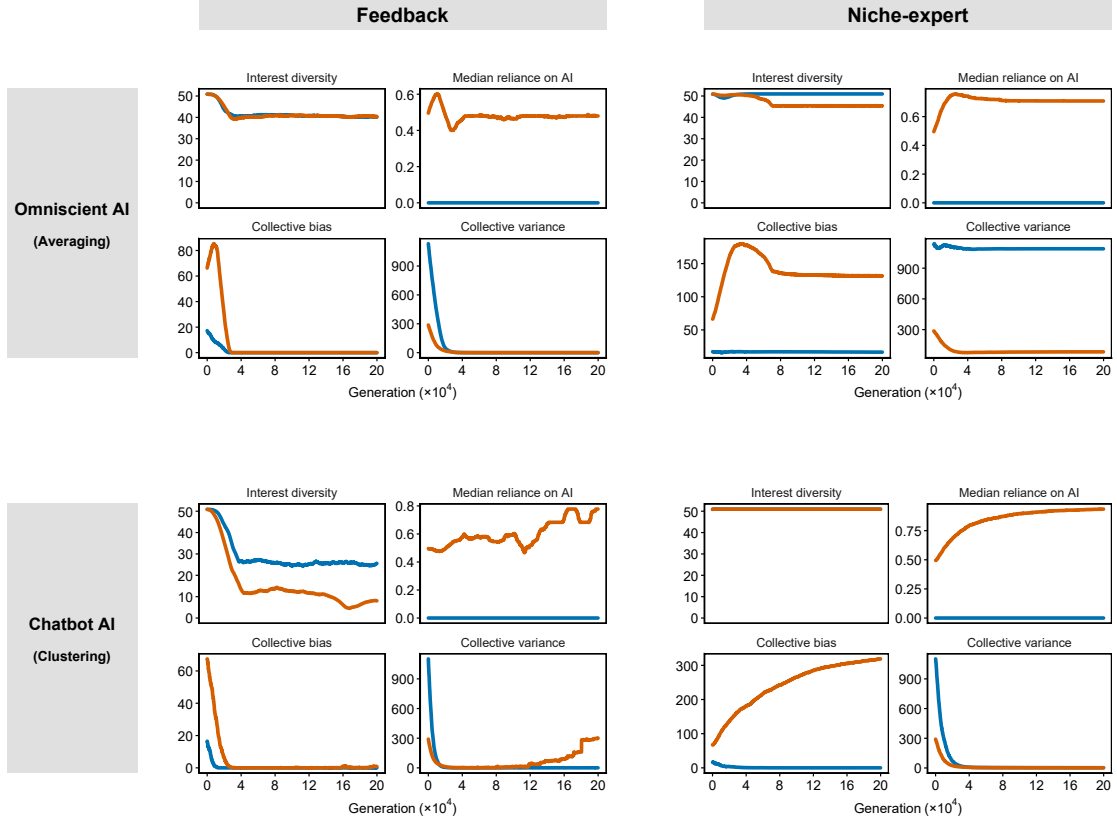


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as hill-shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance repressent variability of the collective prediction error across prediction events. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

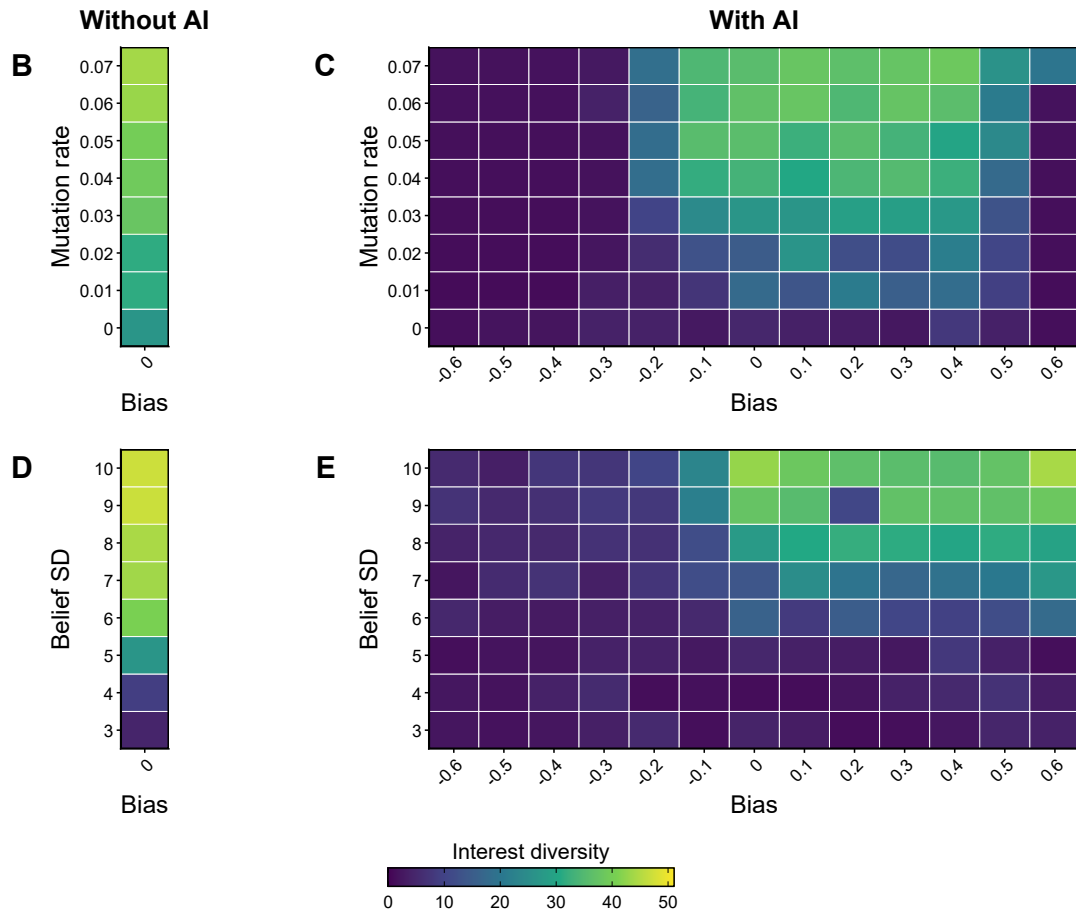
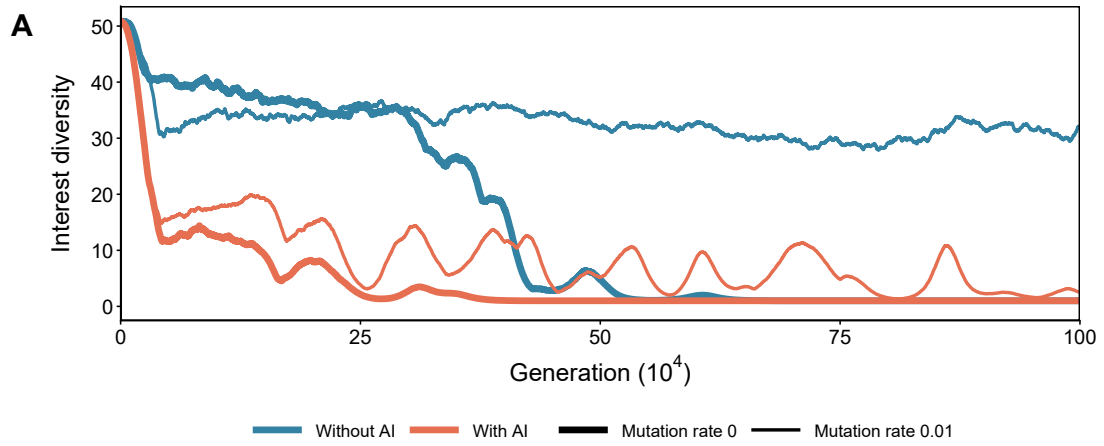


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, chatbot AI is used.

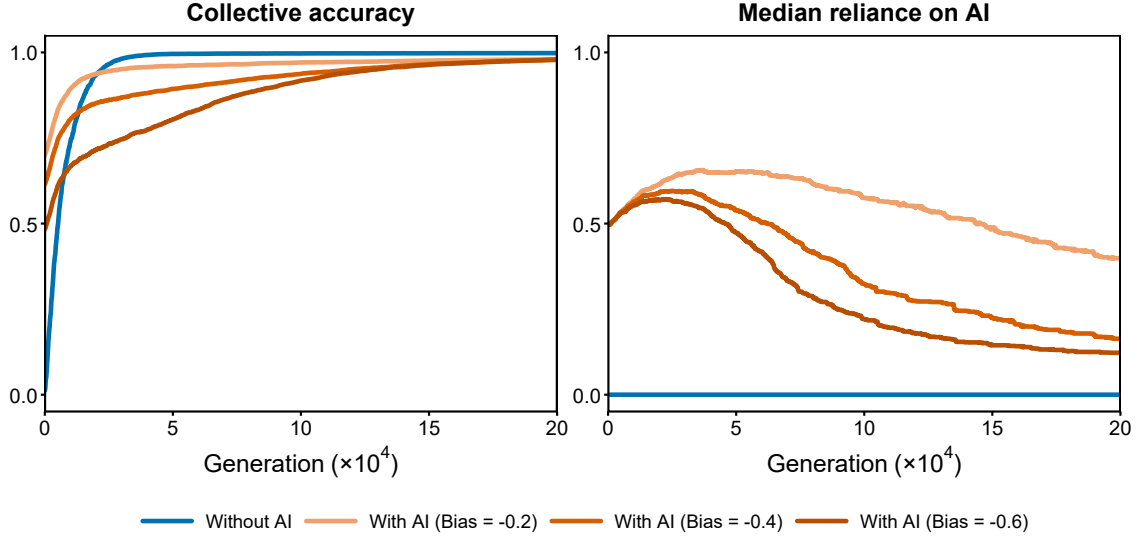


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

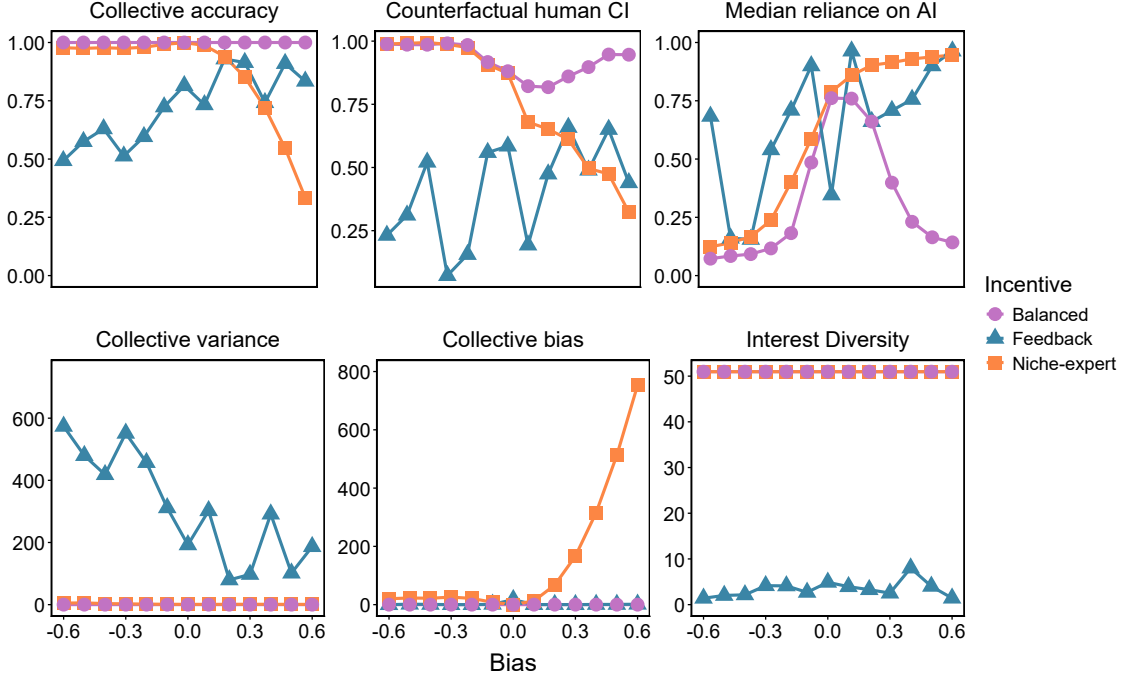


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,000–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

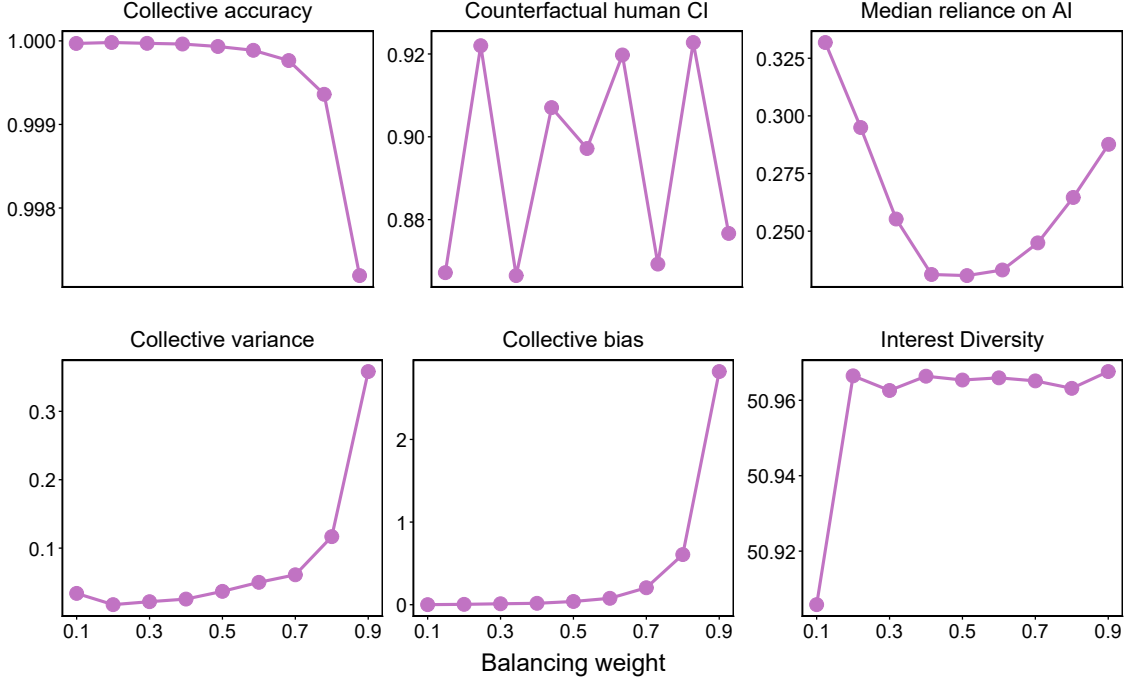


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, chatbot AI is used.

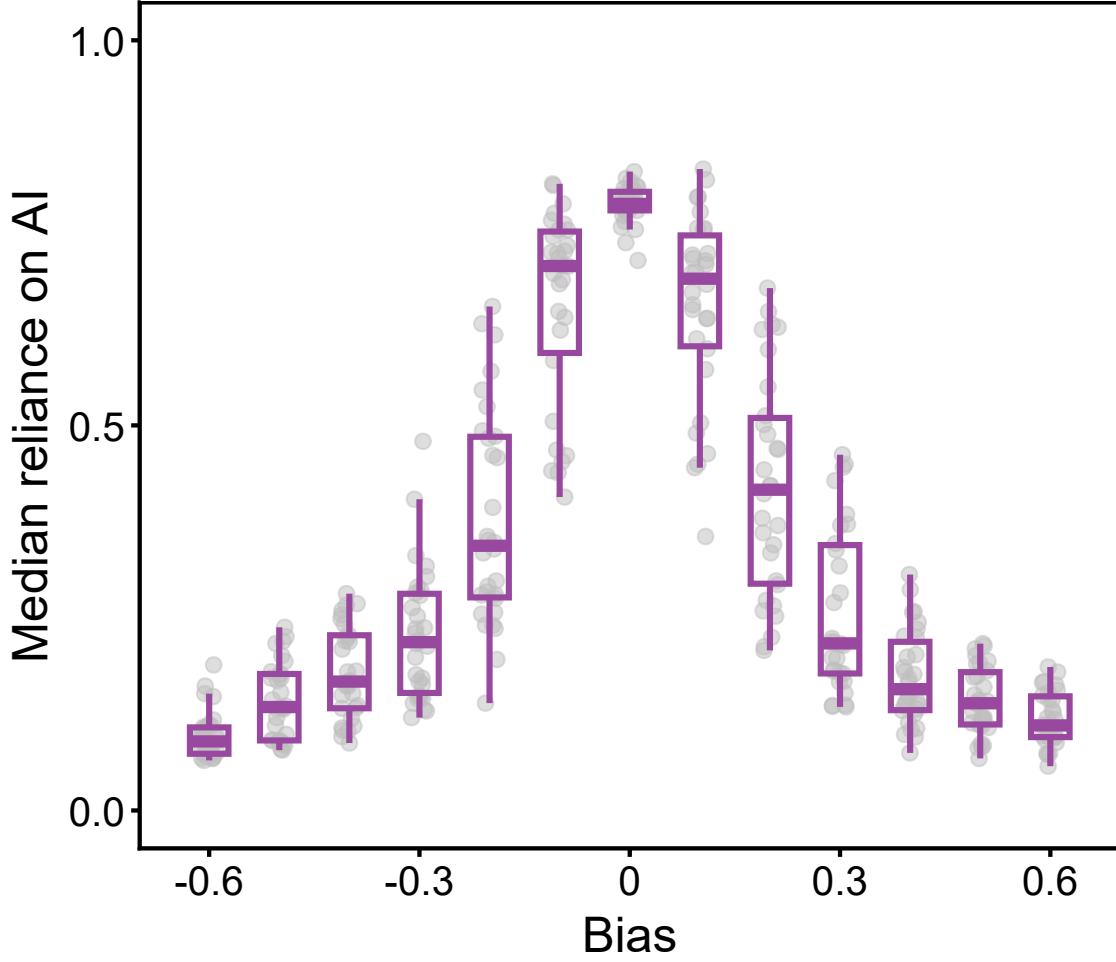


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

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